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MTH622 QUIZ#1 FILE

SOLVED BY SANAULLAH KHAN

CONTACT NO:03066793776

Subject Enrolment

1. Online classes are available (Mathematics)
2. Solved graded activities
   - Assignment's
   - Quizzes
   - GDB's
3. Solved quiz files
4. Past papers
5. Any info about virtual university of Pakistan

Final project (MTH620)

Email:

Sanaullahkhan03066793@gmail.com

Facebook:

Sanaullah Khan

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Quiz

vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=6c56b15e-3ca3-445c-8aa7-d05b86f7c00c

MC200202068: FAIZA QAMAR 52

1917x800 Recording [00:00:13]

MTH622 - Vectors and Classical Mechanics (QUIZ NO.1) **Quiz Start Time: 12:41 PM, 17 February 2022**

Question # 1 of 10 (Start time: 12:41:16 PM, 17 February 2022) **Total Marks: 1**

The frequency f of an oscillatory system is equal to

Select the correct option

Reload Math Equations

- (selected) $f = \frac{1}{T}$
- $f = e^{T}$
- $f = e^{-T}$
- $f = -\frac{1}{T}$

12:41 PM 2/17/2022

Quiz

vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=100e2b64-de88-4563-a968-56a0099c1da8

MTH622 - Vectors and Classical Mechanics (QUIZ NO.1) **Quiz Start Time: 12:41 PM, 17 February 2022**

1917x800 Recording [00:00:30]

Question # 2 of 10 (Start time: 12:42:01 PM, 17 February 2022) **Total Marks: 1**

The force field $\vec{F} = -x\hat{i} - 2z\hat{k}$ is

Select the correct option

Reload Math Equations

- non conservative
- (selected) conservative

Click to Save Answer & Move to Next Question

12:42 PM 2/17/2022

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Quiz

vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=cf0d664b-17c6-422a-8c67-bf9a91e51989

MTH622 - Vectors and Classical Mechanics (QUIZ NO.1)   Quiz Start Time: 12:41 PM, 17 February 2022

1917x800   Recording [00:00:48]

Question # 3 of 10 (Start time: 12:42:21 PM, 17 February 2022)   Total Marks: 1

Given $\vec{F} = \hat{j} + 4\hat{k}$ and $\vec{r} = 2\hat{k} + \hat{j}$, Find the work done W.

Select the correct option

Reload Math Equations

- ○ 6 units
- ● 9 units

Click to Save Answer & Move to Next Question

12:42 PM 2/17/2022

Quiz

vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=b337e1a3-2e29-42db-9f11-07f366da904c

1917x800   Recording [00:01:14]

The trajectory of a particle in projectile motion with air resistance is

Select the correct option

- ○ none of these
- ○ a hyperbola
- ○ a parabola
- ● a non parabolic curve

12:42 PM 2/17/2022

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Forces that cannot be expressed in the term of a potential energy function are called non conservative forces.

Select the correct option

- false
- true

A car driving along a curved road is an example of

Select the correct option

- curvilinear motion
- oscillatory motion
- none of these options
- rectilinear motion

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Quiz

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MTH622 - Vectors and Classical Mechanics (QUIZ NO.1) Quiz Start Time: 12:41 PM, 17 February 2022

1917x800 Recording [00:02:08]

Question # 7 of 10 (Start time: 12:43:34 PM, 17 February 2022) **Total Marks: 1**

Kinematics is the branch of mechanics that deals with the moving objects and the forces which cause the motion.

Select the correct option

- ○ true
- ● false

12:43 PM 2/17/2022

Quiz

vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=cea8e7b9-9cfc-42b9-8ea4-d09f06e8be96

1917x800 Recording [00:03:00]

An object of mass m linked to a spring of spring constant k represents the

Select the correct option

- ○ rectilinear motion
- ● simple harmonic motion

12:44 PM 2/17/2022

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Quiz

vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=e52c6c07-0920-4397-89ed-8a75e379d041

MTH622 - Vectors and Classical Mechanics (QUIZ NO.1) **Quiz Start Time: 12:53 PM, 17 February 2022**

Question # 1 of 10 (Start time: 12:53:44 PM, 17 February 2022) **Total Marks: 1**

The force field $\vec{F} = 3x\hat{i} - 2y\hat{\hat{j}}$ is

Select the correct option

Reload Math Equations

- ( ) non conservative
- (•) conservative

Click to Save Answer & Move to Next Question

12:53 PM 2/17/2022

Quiz

vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=52a829ae-6f1c-4a95-aca8-2490a7cd3051

Question # 2 of 10 (Start time: 12:53:55 PM, 17 February 2022) **Total Marks: 1**

Forces that can be expressed in the term of a potential energy function are called non conservative forces.

Select the correct option

- ( ) true
- (•) false

Click to Save Answer & Move to Next Question

12:54 PM 2/17/2022

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Quiz

vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=ac489f0e-ae89-4899-9501-c63f544d787e

MTH622 - Vectors and Classical Mechanics (QUIZ NO.1) — **Quiz Start Time: 12:53 PM, 17 February 2022**

Question # 3 of 10 (Start time: 12:54:08 PM, 17 February 2022) — **Total Marks: 1**

Find the time period T of the vibrating mass with $m = 5$ and $f = 10$

Select the correct option

Reload Math Equations

- ( ) $T = \frac{1}{2}$
- (•) $T = \frac{1}{10}$

Click to Save Answer & Move to Next Question

12:54 PM 2/17/2022

Quiz

vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=c4702821-fb06-4608-b17e-167528e718b6

Question # 4 of 10 (Start time: 12:54:19 PM, 17 February 2022) — **Total Marks: 1**

The forces acting on a damped harmonic oscillator
tend to .... the amplitude of the successive oscillations.

Select the correct option

- (•) decrease
- ( ) increase

Click to Save Answer & Move to Next Question

12:54 PM 2/17/2022

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Quiz
vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=89b8922a-1ade-4389-acf9-55d7f7c3fc2c

A car driving along a curved road is an example of

Select the correct option

- (selected) curvilinear motion
- none of these options
- rectilinear motion
- oscillatory motion

Click to Save Answer & Move to Next Question

12:54 PM 2/17/2022

Quiz
vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=2bab2893-6b22-4861-8048-ad8774c7d295

Question # 6 of 10 (Start time: 12:54:43 PM, 17 February 2022) **Total Marks: 1**

Kinematics is the branch of mechanics that deals with the moving objects and the forces which cause the motion.

Select the correct option

- (selected) false
- true

Click to Save Answer & Move to Next Question

12:54 PM 2/17/2022

Quiz

vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=a7680d25-14ff-4d38-85d3-6e9dcf50b652

MTH622 - Vectors and Classical Mechanics (QUIZ NO.1) **Quiz Start Time: 12:53 PM, 17 February 2022**

Question # 7 of 10 (Start time: 12:54:56 PM, 17 February 2022) **Total Marks: 1**

If position of a vibrating mass at any time is defined as $y(t) = 20cos(\frac{3}{4}t)$. Then the amplitude is

Select the correct option

Reload Math Equations

- (selected) 20
- $\frac{3}{4}$

Click to Save Answer & Move to Next Question

12:55 PM 2/17/2022

Quiz

vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=fdbc845a-df7d-47b1-9c28-eeea06fdfb62

The damping force of a harmonic oscillator is proportional to its ......

Select the correct option

- (selected) velocity
- displacement
- acceleration
- time

Click to Save Answer & Move to Next Question

12:55 PM 2/17/2022

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Chasle's theorem states that the most general rigid body displacement can be produced by a translation along a line followed by a .... about that line.

Select the correct option

- (selected) rotation
- reflection
- displacement
- translation

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (QUIZ NO.1) **Quiz Start Time: 12:53 PM, 17 February 2022**

Question # 10 of 10 (Start time: 12:55:49 PM, 17 February 2022) **Total Marks: 1**

If position of a vibrating mass at any time is defined as $y(t) = 20cos(\frac{2}{3}t)$. Then the time period is

Select the correct option

Reload Math Equations

- $\frac{3}{2}\pi$
- (selected) 3π

Click to Save Answer & Move to Next Question

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Quiz | (357) WhatsApp

vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=5995b8a7-a4f7-4a55-8033-140c74abe8ff

Question # 1 of 10 (Start time: 02:18:26 PM, 17 February 2022) **Total Marks: 1**

The forces acting on a damped harmonic oscillator
tend to .... the amplitude of the successive oscillations.

Select the correct option

- ○ increase
- ● decrease

Click to Save Answer & Move to Next Question

2:18 PM 2/17/2022

Quiz | (357) WhatsApp

vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=7004dca5-ea7f-4672-84c9-b2cf4b83b6da

A freely falling object observes

Select the correct option

- ○ oscillatory motion
- ● uniformly accelerated motion
- ○ none of these
- ○ uniform rectilinear motion

Click to Save Answer & Move to Next Question

2:18 PM 2/17/2022

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Quiz | (357) WhatsApp

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Question # 3 of 10 (Start time: 02:18:59 PM, 17 February 2022) **Total Marks: 1**

Time period is .... required by a oscillation system to complete its one cycle of oscillation of the specific system.

Select the correct option

- ( ) maximum time
- (•) minimum time

Click to Save Answer & Move to Next Question

2:19 PM 2/17/2022

Quiz | (357) WhatsApp

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Question # 4 of 10 (Start time: 02:19:25 PM, 17 February 2022) **Total Marks: 1**

In case of conservative force field, the total energy is........

Select the correct option

- (•) a constant
- ( ) any function of time

Click to Save Answer & Move to Next Question

2:19 PM 2/17/2022

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The trajectory of a particle in projectile motion with air resistance is

Select the correct option

- ○ a parabola
- ◉ a non parabolic curve
- ○ a hyperbola
- ○ none of these

Click to Save Answer & Move to Next Question

2:19 PM 2/17/2022

Quiz | (357) WhatsApp

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Question # 6 of 10 (Start time: 02:19:49 PM, 17 February 2022) **Total Marks: 1**

Friction is a

Select the correct option

- ○ conservative force
- ◉ non conservative force

Click to Save Answer & Move to Next Question

2:19 PM 2/17/2022

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MTH622 - Vectors and Classical Mechanics (QUIZ NO.1) Quiz Start Time: 02:18 PM, 17 February 2022

Question # 7 of 10 (Start time: 02:20:00 PM, 17 February 2022) **Total Marks: 1**

If the angle between force and displacement is greater than $\frac{\pi}{2}$ then the work done will be

Select the correct option

Reload Math Equations

- (selected) negative
- positive

Click to Save Answer & Move to Next Question

2:20 PM 2/17/2022

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MTH622 - Vectors and Classical Mechanics (QUIZ NO.1) Quiz Start Time: 02:18 PM, 17 February 2022

Question # 8 of 10 (Start time: 02:20:15 PM, 17 February 2022) **Total Marks: 1**

Gravitational force is an example of a conservative force.

Select the correct option

Reload Math Equations

- (selected) true
- false

Click to Save Answer & Move to Next Question

2:20 PM 2/17/2022

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Question # 9 of 10 (Start time: 02:20:27 PM, 17 February 2022) **Total Marks: 1**

Forces that can be expressed in the term of a potential energy function are called non conservative forces.

Select the correct option

- (selected) false
- true

Click to Save Answer & Move to Next Question

2:20 PM 2/17/2022

Quiz | (357) WhatsApp

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MTH622 - Vectors and Classical Mechanics (QUIZ NO.1) **Quiz Start Time: 02:18 PM, 17 February 2022**

Question # 10 of 10 (Start time: 02:20:37 PM, 17 February 2022) **Total Marks: 1**

If position of a vibrating mass at any time is defined as $y(t) = 20cos(\frac{3}{4}t)$. Then the time period will be

Select the correct option

Reload Math Equations

- $\frac{4}{3}\pi$
- (selected) $\frac{8}{3}\pi$

Click to Save Answer & Move to Next Question

2:20 PM 2/17/2022

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Question # 1 of 10 (**Start time: 02:22:36 PM, 17 February 2022**) **Total Marks: 1**

Forces that can be expressed in the term of a potential energy function are called non conservative forces.

Select the correct option

- (selected) false
- true

Click to Save Answer & Move to Next Question

2:22 PM 2/17/2022

Quiz

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MTH622 - Vectors and Classical Mechanics (QUIZ NO.1) **Quiz Start Time: 02:22 PM, 17 February 2022**

Question # 2 of 10 (**Start time: 02:22:47 PM, 17 February 2022**) **Total Marks: 1**

Given $\vec{F} = \hat{j} + 4\hat{k}$ and $\vec{r} = 2\hat{k} + \hat{j}$, Find the work done W.

Select the correct option

Reload Math Equations

- (selected) 9 units
- 6 units

Click to Save Answer & Move to Next Question

2:22 PM 2/17/2022

Quiz

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MTH622 - Vectors and Classical Mechanics (QUIZ NO.1) Quiz Start Time: 02:22 PM, 17 February 2022

Question # 3 of 10 (Start time: 02:22:58 PM, 17 February 2022) **Total Marks: 1**

Given $\vec{F} = 3\hat{i} + 4\hat{j}$ and $\vec{r} = 2\hat{j} - \hat{i}$, Find the work done W.

Select the correct option

Reload Math Equations

- (selected) 5 units
- 2 units

Click to Save Answer & Move to Next Question

2:23 PM 2/17/2022

Quiz

vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=ba118066-6499-4213-8d52-c1b774d4d43c

Question # 4 of 10 (Start time: 02:23:09 PM, 17 February 2022) **Total Marks: 1**

Time period is ...... required by a oscillation system to complete its one cycle of oscillation of the specific system.

Select the correct option

- maximum time
- (selected) minimum time

Click to Save Answer & Move to Next Question

2:23 PM 2/17/2022

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Quiz

vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=9f2bed9b-f4df-4484-b847-50455bd8846c

MTH622 - Vectors and Classical Mechanics (QUIZ NO.1) **Quiz Start Time: 02:22 PM, 17 February 2022**

Question # 5 of 10 (Start time: 02:23:20 PM, 17 February 2022) **Total Marks: 1**

The basis vectors for cylindrical coordinate system are

Select the correct option

Reload Math Equations

- ( ) $e_\phi = \cos\phi\hat{i} + \sin\phi\hat{j}, e_\rho = -\sin\phi\hat{i} + \cos\phi\hat{j}$, and $e_z = \hat{k}$
- (•) $e_\rho = \cos\phi\hat{i} + \sin\phi\hat{j}, e_\phi = -\sin\phi\hat{i} + \cos\phi\hat{j}$, and $e_z = \hat{k}$

Click to Save Answer & Move to Next Question

2:23 PM 2/17/2022

Quiz

vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=f4b4e360-a901-4af4-a0fb-59615529280f

The trajectory of a particle in projectile motion with air resistance is

Select the correct option

- ( ) none of these
- (•) a non parabolic curve
- ( ) a hyperbola
- ( ) a parabola

Click to Save Answer & Move to Next Question

2:23 PM 2/17/2022

Quiz

vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=cbed452d-e69d-4d68-9446-4968296dcd22

Question # 7 of 10 (Start time: 02:23:39 PM, 17 February 2022) **Total Marks: 1**

Kinematics is the branch of mechanics that deals with the moving objects and the forces which cause the motion.

Select the correct option

- (selected) false
- true

Click to Save Answer & Move to Next Question

2:23 PM 2/17/2022

Quiz

vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=5ec226b2-3c33-4908-a1d6-16d767b18f2d

The damping force of a harmonic oscillator is proportional to its ......

Select the correct option

- acceleration
- displacement
- time
- (selected) velocity

Click to Save Answer & Move to Next Question

2:23 PM 2/17/2022

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MTH622 - Vectors and Classical Mechanics (QUIZ NO.1) **Quiz Start Time: 02:22 PM, 17 February 2022**

Question # 9 of 10 (Start time: 02:24:02 PM, 17 February 2022) **Total Marks: 1**

If the angle between force and displacement is $\frac{\pi}{2}$ then the work done will be.

Select the correct option

Reload Math Equations

- (selected) zero
- maximum

Click to Save Answer & Move to Next Question

2:24 PM 2/17/2022

Quiz

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Question # 10 of 10 (Start time: 02:24:33 PM, 17 February 2022) **Total Marks: 1**

In case of conservative force field, the total energy is.......

Select the correct option

- any function of time
- (selected) a constant

Click to Save Answer & Move to Next Question

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